

Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence

Erik Brynjolfsson* Bharat Chandar† Ruyu Chen‡§¶

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Abstract

This paper examines changes in the labor market for occupations exposed to generative artificial intelligence using high-frequency administrative data from the largest payroll software provider in the United States. We present six facts that characterize these shifts. We find that since the widespread adoption of generative AI, early-career workers (ages 22-25) in the most AI-exposed occupations have experienced a 13 percent relative decline in employment even after controlling for firm-level shocks. In contrast, employment for workers in less exposed fields and more experienced workers in the same occupations has remained stable or continued to grow. We also find that adjustments occur primarily through employment rather than compensation. Furthermore, employment declines are concentrated in occupations where AI is more likely to *automate*, rather than *augment*, human labor. Our results are robust to alternative explanations, such as excluding technology-related firms and excluding occupations amenable to remote work. These six facts provide early, large-scale evidence consistent with the hypothesis that the AI revolution is beginning to have a significant and disproportionate impact on entry-level workers in the American labor market.

*Stanford University and NBER; erikb@stanford.edu

†Stanford University; chandarb@stanford.edu

‡Stanford University; ruyuchen@stanford.edu

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- Fifth, these labor market adjustments are more visible in employment than in compensation.
- Sixth, we find that these patterns hold in occupations unaffected by remote work and across various alternative sample constructions.

While our main estimates may be influenced by factors other than generative AI, our results are consistent with the hypothesis that generative AI has begun to significantly affect entry-level employment.

The adoption of new technologies typically leads to heterogeneous effects across workers, resulting in an adjustment period as workers reallocate from displaced forms of work to new forms with growing labor demand (Autor et al., 2024). Such endogenous adjustment may already be happening with AI, with emerging evidence of shifts in college majors away from AI-exposed categories such as computer science (Horowitz, 2025). Past transitions such as the IT revolution ultimately led to robust growth in employment and real wages following physical and human capital adjustments, with some workers benefiting more than others (Bresnahan et al., 2002; Brynjolfsson et al., 2021).

Tracking employment trends on an ongoing basis will help determine if the adjustment to AI follows a similar pattern. Consequently, we will continue to monitor these outcomes to assess whether the trends documented in the paper accelerate in the future. Future work would benefit from better firm-level AI adoption data, which would provide sharper variation for estimating plausible causal effects of AI on employment.

References

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